

Imbalance conjecture

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id imbalance_conjecture — generated 2026-09-16

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Summary and provenance

Catalog id	imbalance_conjecture
Catalog entry	Imbalance conjecture
Source corpus	opg
Campaign leg	OpenProblemGarden first pass (attacks_opg)
Source paper / entry	http://www.openproblemgarden.org/op/imbalance_conjecture
Model verdict	proved (confidence: high)
Model's one-line claim	A truncated-tail inequality for edge imbalances implies every Erdős–Gallai inequality and proves the conjecture.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_opg/imbalance_conjecture/output.md
Independent review	None
Model's caveats	The argument is self-contained apart from the standard Erdős–Gallai criterion; publication novelty has not been checked.

Problem statement

Conjecture Suppose that for all edges $e \in E(G)$ we have $imb(e) > 0$. Then M_G is graphic.

Catalog context

Consider simple undirected graph G and let $e = uv \in E(G)$. The imbalance of the edge e defined as $imb(e) = |deg(u) - deg(v)|$. The multiset of all edge imbalances of G is denoted by M_G . Note, that conjecture is verified for all such graphs with ≤ 9 vertices.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

Theorem

Let G be a finite simple undirected graph such that adjacent vertices have different degrees. Then

$$M_G = \{|d(u) - d(v)| : uv \in E(G)\}$$

is the degree multiset of a simple graph.

Write

$$a_e = |d(u) - d(v)| \quad (e = uv), \quad m = |E(G)|, \quad D = \Delta(G).$$

The hypothesis is $a_e \geq 1$ for every edge.

The main ingredient is the following inequality.

Truncated-tail lemma

If $m > 0$, then for every integer $1 \leq k < D$ and every set $S \subseteq E(G)$ of k edges,

$$\sum_{e \in E(G) \setminus S} \min\{k, a_e\} \geq k(D - k). \quad (1)$$

Proof First, $m \geq D$, and the hypothesis implies $D \geq 2$.

For $k = 1$, positivity gives

$$\sum_{e \notin S} \min\{1, a_e\} = m - 1 \geq D - 1,$$

as required.

We also dispose of $D = 3$, $k = 2$. If $m \geq 4$, positivity gives a lower bound $m - 2 \geq 2$. If $m = 3$, the degree-three vertex is incident with every edge, so G consists of $K_{1,3}$ and isolated vertices. All three imbalances are 2, and (1) again holds. These observations cover all cases with $D \leq 3$.

Hence assume $D \geq 4$ and $2 \leq k < D$.

Fix a vertex v of degree D . Let

$$q = m - D$$

be the number of edges not incident with v , and put

$$s = |S \cap E(v)|, \quad t = |S \setminus E(v)| = k - s.$$

Define

$$U = \{u \in N(v) : vu \notin S\}, \quad r = |U| = D - s.$$

Finally, put

$$b = D - k.$$

Thus $b \geq 1$, $0 \leq t \leq k$, and

$$r = b + t. \quad (2)$$

There are exactly $q - t$ edges outside S that are not incident with v . Each contributes at least 1 to the left side of (1). Moreover, since v has maximum degree,

$$a_{vu} = D - d(u).$$

Consequently,

$$\begin{aligned} \sum_{e \notin S} \min\{k, a_e\} &\geq q - t + \sum_{u \in U} \min\{k, D - d(u)\} \\ &= q - t + kr - \sum_{u \in U} (d(u) - b)_+, \end{aligned} \quad (3)$$

where $x_+ = \max\{x, 0\}$.

Let

$$L = \{u \in U : d(u) > b\}, \quad \ell = |L|.$$

Because every vertex of L is adjacent to v ,

$$\sum_{u \in L} (d(u) - 1)$$

counts incidences of vertices in L with edges not incident with v . Such an edge is counted at most once, except that an edge with both endpoints in L is counted twice. Simplicity therefore gives

$$\sum_{u \in L} (d(u) - 1) \leq q + |E(G[L])| \leq q + \binom{\ell}{2}.$$

It follows that

$$\sum_{u \in U} (d(u) - b)_+ = \sum_{u \in L} (d(u) - b) \leq q + \binom{\ell}{2} - (b-1)\ell. \quad (4)$$

Set

$$F_b(\ell) = \binom{\ell}{2} - (b-1)\ell.$$

Combining (3) and (4), we obtain

$$\sum_{e \notin S} \min\{k, a_e\} \geq kr - t - F_b(\ell). \quad (5)$$

We claim that

$$F_b(\ell) \leq (k-1)t. \quad (6)$$

Indeed, $0 \leq \ell \leq r = b + t$, and $F_b(x)$ is a convex quadratic. Its maximum on this interval is attained at an endpoint. Hence

$$\begin{aligned} F_b(\ell) &\leq \max\{F_b(0), F_b(b+t)\} \\ &= \max\left\{0, \binom{t+1}{2} - \binom{b}{2}\right\}. \end{aligned} \quad (7)$$

If $k \geq 3$, then

$$\max\left\{0, \binom{t+1}{2} - \binom{b}{2}\right\} \leq \binom{t+1}{2} \leq \frac{t(k+1)}{2} \leq (k-1)t.$$

If $k = 2$, then $b = D - 2 \geq 2$ and $t \in \{0, 1, 2\}$. The last expression in (7) is respectively at most 0, 0, 2, and hence is at most $t = (k-1)t$. This proves (6).

Using (6) in (5), and then (2), gives

$$\sum_{e \notin S} \min\{k, a_e\} \geq kr - t - (k-1)t = k(r-t) = kb = k(D-k).$$

This proves the lemma. $\square$

Completion of the proof

If $m = 0$, the multiset is empty and is graphic. Assume $m > 0$.

First, the imbalance sum is even:

$$\begin{aligned} \sum_{uv \in E(G)} |d(u) - d(v)| &\equiv \sum_{uv \in E(G)} (d(u) + d(v)) \\ &= \sum_{v \in V(G)} d(v)^2 \\ &\equiv \sum_{v \in V(G)} d(v) = 2m \equiv 0 \pmod{2}. \end{aligned} \quad (8)$$

Also, both endpoints of an edge have positive degree, so

$$1 \leq a_e \leq D - 1. \quad (9)$$

Order the imbalances as

$$a_1 \geq a_2 \geq \cdots \geq a_m.$$

By the Erdős–Gallai criterion, it remains to prove, for every $1 \leq k \leq m$,

$$\sum_{i=1}^k a_i \leq k(k-1) + \sum_{i=k+1}^m \min\{k, a_i\}. \quad (10)$$

If $k \geq D$, then (9) gives

$$\sum_{i=1}^k a_i \leq k(D-1) \leq k(k-1),$$

so (10) holds.

If $1 \leq k < D$, apply the truncated-tail lemma to a set of edges carrying the k largest imbalances. It gives

$$\sum_{i=k+1}^m \min\{k, a_i\} \geq k(D-k).$$

Therefore,

$$\begin{aligned} k(k-1) + \sum_{i=k+1}^m \min\{k, a_i\} &\geq k(k-1) + k(D-k) \\ &= k(D-1) \\ &\geq \sum_{i=1}^k a_i. \end{aligned}$$

Thus every Erdős–Gallai inequality holds. Together with (8), this proves that M_G is graphic. $\square$

Scope and verification status

The argument covers disconnected graphs, isolated vertices, the empty graph, and all maximum degrees. Simplicity is used explicitly in the estimate

$$|E(G[L])| \leq \binom{|L|}{2},$$

and positive imbalance is used to obtain one unit of tail contribution from every retained edge not incident with the chosen maximum-degree vertex.

No conjectural auxiliary assertion or computational check is needed. I have not independently checked whether this proof has appeared in the literature.